



Customer Shopping Behavior Analysis

Insights from 3,900 purchases using Python, SQL and Power BI

3,900 purchases, three tools, one goal

Understand spending patterns, segments, product preferences, discounts, shipping and subscriptions.

3,900

purchases

18

columns

37

missing ratings

Python

Preparation & exploration

SQL

Business analysis

Power BI

Dashboard & communication

Python cleaned and enriched the data

Dataset: demographics, purchase details and shopping behavior across 18 columns.

01

Impute

Filled missing ratings with each category's median

02

Standardize

Renamed columns to snake_case

03

Engineer

Added age_group and purchase_frequency_days

04

Simplify

Dropped promo_code_used, redundant with discount_applied

05

Load

Loaded cleaned data into the database for SQL

Young Adults and men drive revenue

REVENUE BY AGE GROUP

1 Young Adults

2 Middle-aged

3 Adult

4 Senior

BY GENDER

Male

customers generate the larger total revenue; female customers remain a material share.

Loyal customers rarely subscribe

SEGMENTS BY PREVIOUS PURCHASES

Loyal	Largest segment
Returning	Second
New	Smallest

SUBSCRIPTION (5+ PRIOR PURCHASES)

	Subscribers	Non-subs
Avg. spend	Slightly lower	Higher
Group size	Smaller	Larger
Total revenue	Lower	Much higher

Loyal behavior is not yet converting into subscriptions: non-subscribers outnumber subscribers among repeat buyers.

Discounts and shipping: who spends more

839

customers used a discount
and still spent above average

Discounts are not limited to low-value orders.

50%

of hat purchases involved a
discount, the highest
incidence of any product

Hats lead discount dependence.

Express

shipping carries a higher
average purchase amount
than Standard

*A premium segment worth
targeting.*

One dashboard brings it together

KPI cards

- Number of customers
- Average purchase amount
- Average review rating

Charts

- Subscription status
- Revenue & sales by category
- Revenue & sales by age group

Filters

- Subscription status
- Gender
- Category
- Shipping type

Five recommendations

- | | | |
|-----------|-------------------------------|--|
| 01 | Boost subscriptions | Exclusive benefits and clearer value |
| 02 | Build loyalty programs | Reward repeat buyers |
| 03 | Review discount policy | Balance sales and margin |
| 04 | Feature top products | Promote top-rated and best-selling items |
| 05 | Target key customers | High-revenue age groups and Express shippers |

Python prepares the data, SQL supplies the evidence, Power BI keeps it visible.